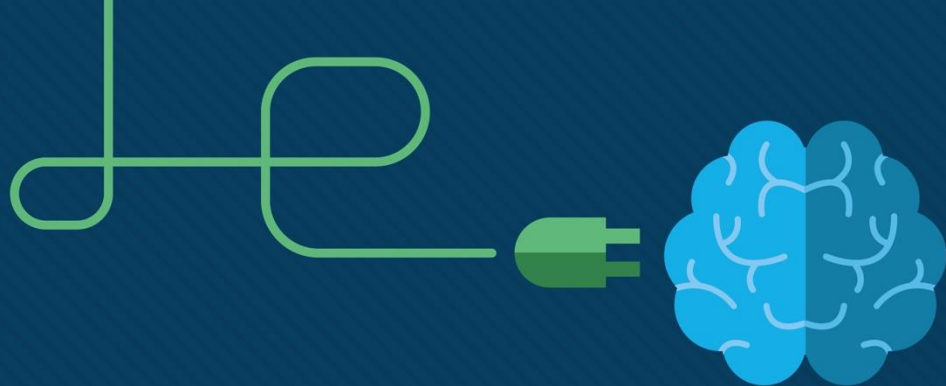




Build Basic LAN Configuration

Introduction to Networks v7.0
(ITN)



Basic Device Configuration

Device Names

- The first configuration command on any device should be to give it a unique hostname.
- By default, all devices are assigned a factory default name. For example, a Cisco IOS switch is "Switch."
- Guideline for naming devices:
 - Start with a **letter**
 - Contain **no spaces**
 - End with a letter or digit
 - Use only **letters**, **digits**, and **dashes**
 - Be **less than 64 characters** in length

```
Switch# configure terminal
Switch(config)# hostname Sw-Floor-1
Sw-Floor-1(config)#
```

Note: To return the switch to the default prompt, use the **no hostname** global config command.

Password Guidelines

- The use of weak or easily guessed passwords are a security concern.
- All networking devices should limit administrative access by securing **privileged EXEC**, **user EXEC**, and **remote Telnet access with passwords**. In addition, all passwords should be encrypted and legal notifications provided.
- Password Guidelines:
 - Use passwords that are **more than eight characters** in length.
 - Use a combination of **upper** and **lowercase letters**, **numbers**, **special characters**, and/or **numeric sequences**.
 - Avoid using the same password for all devices.
 - Do not use common words because they are easily guessed.



Note: Most of the labs in this course use simple passwords such as **cisco** or **class**. These passwords are considered weak and easily guessable and should be avoided in production environments.

Basic Device Configuration

Configure Passwords

Securing user EXEC mode access:

- First enter line console configuration mode using the **line console 0** command in global configuration mode.
- Next, specify the user EXEC mode password using the **password** *password* command.
- Finally, enable user EXEC access using the **login** command.

```
Sw-Floor-1# configure terminal
Sw-Floor-1(config)# line console 0
Sw-Floor-1(config-line)# password cisco
Sw-Floor-1(config-line)# login
Sw-Floor-1(config-line)# end
Sw-Floor-1#
```

Securing privileged EXEC mode access:

- First enter global configuration mode.
- Next, use the **enable secret** *password* command.

```
Sw-Floor-1# configure terminal
Sw-Floor-1(config)# enable secret class
Sw-Floor-1(config)# exit
Sw-Floor-1#
```

Configure Passwords (Cont.)

Securing VTY line access:

- First enter line VTY configuration mode using the **line vty 0 15** command in global configuration mode.
- Next, specify the VTY password using the **password *password*** command.
- Finally, enable VTY access using the **login** command.

```
Sw-Floor-1# configure terminal
Sw-Floor-1(config)# line vty 0 15
Sw-Floor-1(config-line)# password cisco
Sw-Floor-1(config-line)# login
Sw-Floor-1(config-line)# end
Sw-Floor-1#
```

- Note: VTY lines enable remote access using Telnet or SSH to the device. Many Cisco switches support up to **16 VTY lines** that are numbered **0 to 15**.

Basic Device Configuration

Encrypt Passwords

- The startup-config and running-config files display most passwords in plaintext.
- To encrypt all plaintext passwords, use the **service password-encryption** global config command.

```
Sw-Floor-1# configure terminal
Sw-Floor-1(config)# service password-encryption
Sw-Floor-1(config)# exit
Sw-Floor-1#
```

- Use the **show running-config** command to verify that the passwords on the device are now encrypted.

```
Sw-Floor-1# show running-config
!
!
line con 0
password 7 094F471A1A0A
login
!
Line vty 0 4
Password 7 03095A0F034F38435B49150A1819
Login
!
!
end
```

Basic Device Configuration

Banner Messages

- A banner message is important to warn unauthorized personnel from attempting to access the device.
- To create a banner message of the day on a network device, use the **banner motd** *# the message of the day* # global config command.

Note: The “#” in the command syntax is called the delimiting character. It is entered before and after the message.

```
Sw-Floor-1# configure terminal
Sw-Floor-1(config)# banner motd #Authorized Access Only!#
```

The banner will be displayed on attempts to access the device.



```
Press RETURN to get started.
```

```
Authorized Access Only!
```

```
User Access Verification
```

```
Password:
```

Video – Secure Administrative Access to a Switch (Refer 2.4.6)

This video will cover the following:

- Access the command line to secure the switch
- Secure access to the console port
- Secure virtual terminal access for remote access
- Encrypt passwords on the switch
- Configure the banner message
- Verify security changes

2.5 Save Configurations

Save Configurations

Configuration Files

- There are two system files that store the device configuration:
 - startup-config** - This is the saved configuration file that is stored in **NVRAM**. It contains all the commands that will be used by the device upon **startup or reboot**. **Flash does not lose its contents** when the device is **powered off**.
 - running-config** - This is stored in Random Access Memory (**RAM**). It reflects the current configuration. Modifying a running configuration **affects the operation of a Cisco device immediately**. RAM is volatile memory. It **loses all** of its content when the device is **powered off or restarted**.
 - To save changes made to the running configuration to the startup configuration file, use the **copy running-config startup-config** privileged EXEC mode command.

```
Router#show startup-config
Using 624 bytes
!
version 15.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
```

```
Router#show running-config
Building configuration...

Current configuration : 624 bytes
!
version 15.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
```

Alter the Running Configurations

If changes made to the running config do not have the desired effect and the running-config has not yet been saved, you can restore the device to its previous configuration. To do this you can:

- **Remove** the changed **commands individually**.
- **Reload the device** using the **reload** command in privilege EXEC mode. *Note: This will cause the device to briefly go offline, leading to network downtime.*

If the undesired changes were saved to the startup-config, it may be necessary to clear all the configurations using the **erase startup-config** command in privilege EXEC mode.

- **After** erasing the startup-config, **reload the device to clear the running-config file from RAM.**

```
Router# reload
Proceed with reload? [confirm]
Initializing Hardware ...
```

```
Router# erase startup-config
Erasing the nvram filesystem will remove all configuration files! Continue? [confirm]
[OK]
Erase of nvram: complete
%SYS-7-NV_BLOCK_INIT: Initialized the geometry of nvram
Router#
```

Video – Alter the Running Configuration (Refer 2.5.3)

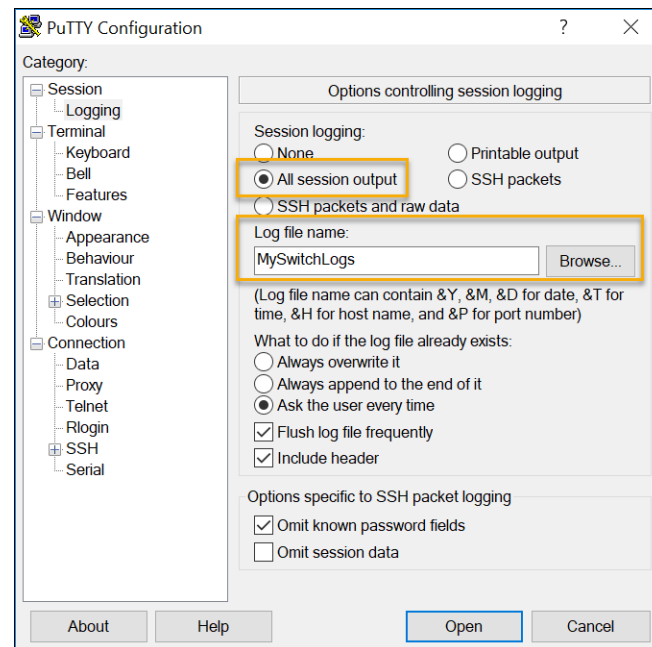
This video will cover the following:

- Copy the running-config file to the startup-config file
- Show the files in the flash or NVRAM directory
- Use command shortening
- Erase the startup-config file
- Copy the start-config file to the running-config file

Capture Configuration to a Text File

Configuration files can also be saved and archived to a text document.

- **Step 1.** Open terminal emulation software, such as PuTTY or Tera Term, that is already connected to a switch.
- **Step 2.** Enable logging in to the terminal software and assign a name and file location to save the log file. The figure displays that **All session output** will be captured to the file specified (i.e., MySwitchLogs).

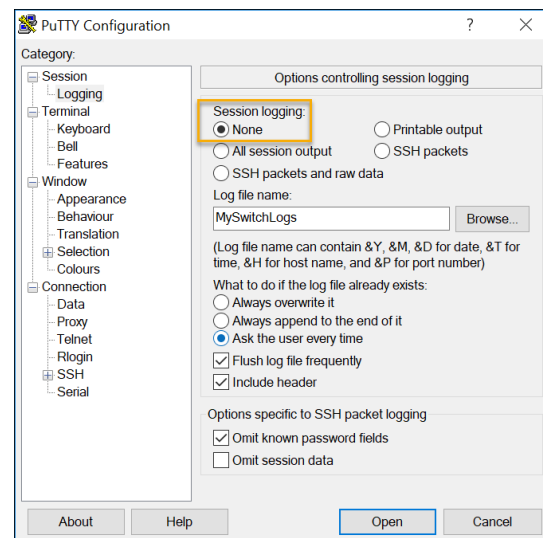


Capture Configuration to a Text File (Cont.)

- **Step 3.** Execute the **show running-config** or **show startup-config** command at the privileged EXEC prompt. Text displayed in the terminal window will be placed into the chosen file.
- **Step 4.** Disable logging in the terminal software. The figure shows how to disable logging by choosing the **None** session logging option

Note: The text file created can be used as a record of how the device is currently implemented. The file could require editing before being used to restore a saved configuration to a device.

```
Switch# show running-config
Building configuration...
```



LabAct 3.1A – Configure Initial Switch Settings

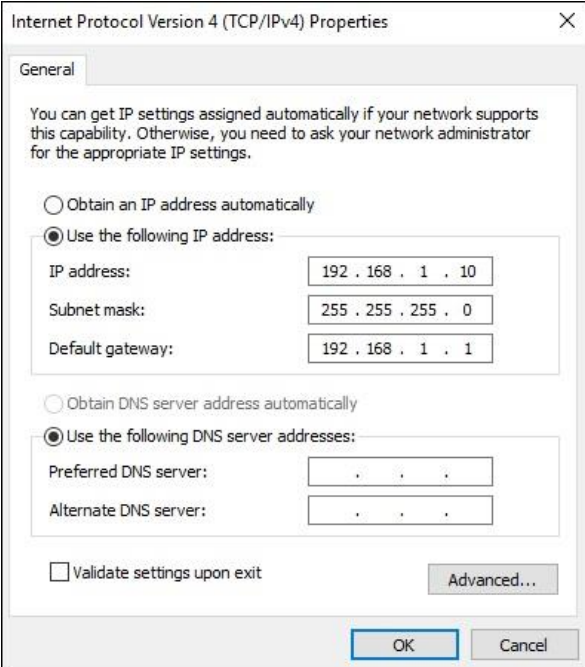
In this Packet Tracer, you will do the following:

- Verify the Default Switch Configuration
- Configure a Basic Switch Configuration
- Configure a MOTD Banner
- Save Configuration Files to NVRAM
- Configure a second Switch

2.6 Ports and Addresses

IP Addresses

- The use of IP addresses is the primary means of enabling devices to locate one another and establish end-to-end communication on the internet.
- The structure of an IPv4 address is called dotted decimal notation and is represented by four decimal numbers between 0 and 255.
- An IPv4 subnet mask is a **32-bit** value that differentiates the network portion of the address from the host portion. Coupled with the IPv4 address, the subnet mask determines to which subnet the device is a member.
- The default gateway address is the IP address of the router that the host will use to access remote networks, including the internet.



The screenshot shows the 'Internet Protocol Version 4 (TCP/IPv4) Properties' dialog box with the 'General' tab selected. The dialog box contains the following elements:

- Title Bar:** 'Internet Protocol Version 4 (TCP/IPv4) Properties' with a close button (X).
- General Tab:** The active tab, showing instructions and configuration options.
- Instructions:** 'You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.'
- IP Address Configuration:**
 - ☐ Obtain an IP address automatically
 - ☒ Use the following IP address:
 - IP address: 192 . 168 . 1 . 10
 - Subnet mask: 255 . 255 . 255 . 0
 - Default gateway: 192 . 168 . 1 . 1
- DNS Configuration:**
 - ☐ Obtain DNS server address automatically
 - ☒ Use the following DNS server addresses:
 - Preferred DNS server: . . .
 - Alternate DNS server: . . .
- ☐ Validate settings upon exit
- Buttons:** 'Advanced...', 'OK', and 'Cancel'.

IP Addresses (Cont.)

- IPv6 addresses are **128 bits** in length and written as a string of **hexadecimal** values. Every four bits is represented by a single hexadecimal digit; for a total of 32 hexadecimal values. Groups of four hexadecimal digits are separated by a colon “.”.

- IPv6 addresses are not case-sensitive and can be written in either lowercase or uppercase.

Note: IP in this course refers to both the IPv4 and IPv6 protocols. IPv6 is the most recent version of IP and is replacing the more common IPv4.

Internet Protocol Version 6 (TCP/IPv6) Properties

General

You can get IPv6 settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IPv6 settings.

☐ Obtain an IPv6 address automatically

☒ Use the following IPv6 address:

IPv6 address: 2001:db8:acad:10::10

Subnet prefix length: 64

Default gateway: fe80::1

☐ Obtain DNS server address automatically

☒ Use the following DNS server addresses:

Preferred DNS server:

Alternate DNS server:

☐ Validate settings upon exit

Advanced...

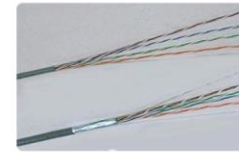
OK Cancel

Interfaces and Ports

- **Network communications** depend on **end user device interfaces**, **networking device interfaces**, and the **cables** that connect them.
- Types of **network media** include **twisted-pair copper cables**, **fiber-optic cables**, **coaxial cables**, or **wireless**.
- Different types of network media have different features and benefits. Some of the differences between various types of media include:
 - **Distance** the media can successfully carry a signal
 - **Environment** in which the media is to be installed
 - **Amount of data** and the **speed** at which it must be transmitted
 - **Cost** of the media and installation



Copper



Fiber-optics



Wireless



2.7 Configure IP Addressing

Manual IP Address Configuration for End Devices

- End devices on the network need an IP address in order to communicate with other devices on the network.
- IPv4 address information can be entered into end devices manually, or automatically using Dynamic Host Configuration Protocol (DHCP).
- To manually configure an IPv4 address on a Windows PC, open the **Control Panel > Network Sharing Center > Change adapter settings** and choose the adapter. Next right-click and select **Properties** to display the **Local Area Connection Properties**.
- Next, click **Properties** to open the **Internet Protocol Version 4 (TCP/IPv4) Properties** window. Then configure the IPv4 address and subnet mask information, and default gateway.

Internet Protocol Version 4 (TCP/IPv4) Properties

General

You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

☐ Obtain an IP address automatically

☒ Use the following IP address:

IP address: 192 . 168 . 1 . 10

Subnet mask: 255 . 255 . 255 . 0

Default gateway: 192 . 168 . 1 . 1

☐ Obtain DNS server address automatically

☒ Use the following DNS server addresses:

Preferred DNS server: . . .

Alternate DNS server: . . .

☐ Validate settings upon exit

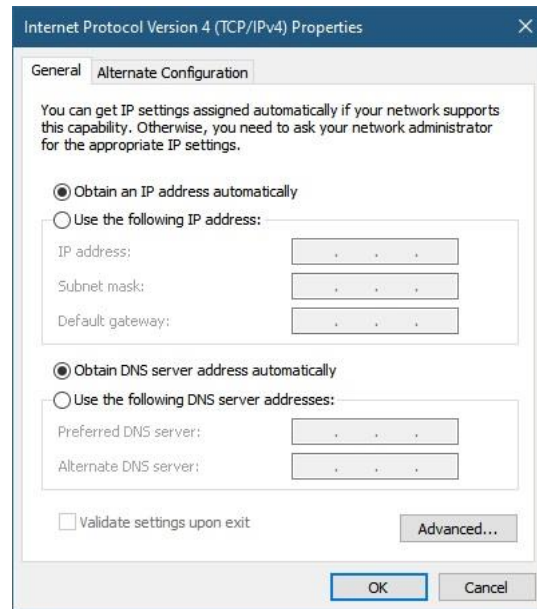
Advanced...

OK Cancel

Note: IPv6 addressing and configuration options are similar to IPv4.

Automatic IP Address Configuration for End Devices

- DHCP enables automatic IPv4 address configuration for every end device that is DHCP-enabled.
- End devices are typically by default using DHCP for automatic IPv4 address configuration.
- To configure DHCP on a Windows PC, open the **Control Panel > Network Sharing Center > Change adapter settings** and choose the adapter. Next right-click and select **Properties** to display the **Local Area Connection Properties**.
- Next, click **Properties** to open the **Internet Protocol Version 4 (TCP/IPv4) Properties** window, then select **Obtain an IP address automatically** and **Obtain DNS server address automatically**.



Note: IPv6 uses DHCPv6 and SLAAC (Stateless Address Autoconfiguration) for dynamic address allocation.

Switch Virtual Interface Configuration

To access the switch remotely, an IP address and a subnet mask must be configured on the SVI.

To configure an SVI on a switch:

- Enter the **interface vlan 1** command in global configuration mode.
- Next assign an IPv4 address using the **ip address** *ip-address subnet-mask command*.
- Finally, enable the virtual interface using the **no shutdown** command.

```
Switch# configure terminal
Switch(config)# interface vlan 1
Switch(config-if)# ip address 192.168.1.20 255.255.255.0
Switch(config-if)# no shutdown
```

Lab3.2A– Basic Device Configuration

In this Packet Tracer, you will do the following:

- Complete the network documentation.
- Perform basic device configurations on a router and a switch.
- Verify connectivity and troubleshoot any issues.

2.8 Verify Connectivity

Video – Test the Interface Assignment refer 2.8.1

This video will cover the following:

- Connect a console cable from the PC to the switch
- Use the terminal emulation program and accept the defaults to bring you to the command line
- Use enable to enter privileged EXEC mode
- Use the global configuration mode and the interface configuration mode to enter the no shutdown command

Video – Test End-to-End Connectivity refer 2.8.2

This video will cover the use of the ping command to test connectivity on both switches and both PCs.

Configure Initial Router Settings

Configure Initial Router Settings

Basic Router Configuration Steps

- Configure the device name.
- Secure privileged EXEC mode.
- Secure user EXEC mode.
- Secure remote Telnet / SSH access.
- Encrypt all plaintext passwords.
- Provide legal notification and save the configuration.

```
Router(config)# hostname hostname
```

```
Router(config)# enable secret password
```

```
Router(config)# line console 0  
Router(config-line)# password password  
Router(config-line)# login
```

```
Router(config)# line vty 0 4  
Router(config-line)# password password  
Router(config-line)# login  
Router(config-line)# transport input {ssh | telnet}
```

```
Router(config)# service password encryption
```

```
Router(config)# banner motd # message #  
Router(config)# end  
Router# copy running-config startup-config
```

Configure Initial Router Settings

Basic Router Configuration Example

- Commands for basic router configuration on R1.
- Configuration is saved to NVRAM.

```
R1(config)# hostname R1
R1(config)# enable secret class
R1(config)# line console 0
R1(config-line)# password cisco
R1(config-line)# login
R1(config-line)# line vty 0 4
R1(config-line)# password cisco
R1(config-line)# login
R1(config-line)# transport input ssh telnet
R1(config-line)# exit
R1(config)# service password encryption
R1(config)# banner motd #
Enter TEXT message. End with a new line and the #
*****
WARNING: Unauthorized access is prohibited!
*****
R1(config)# exit
R1# copy running-config startup-config
```

Lab 3.2B— Configure Initial Router Settings

In this Packet Tracer, you will do the following:

- Verify the default router configuration.
- Configure and verify the initial router configuration.
- Save the running configuration file.

Configure Interfaces

Configure Interfaces

Configure Router Interfaces

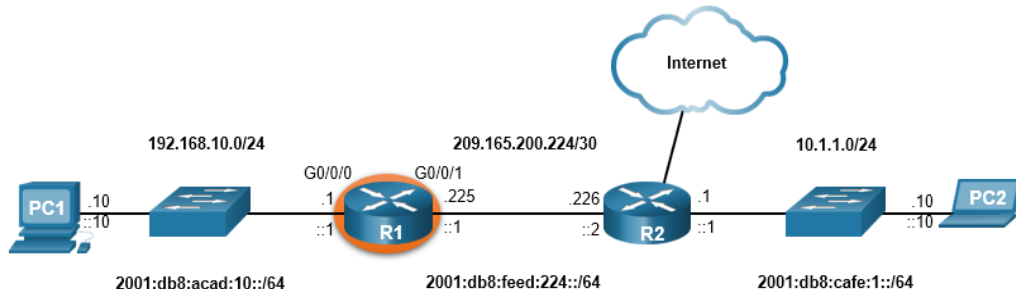
Configuring a router interface includes issuing the following commands:

```
Router(config)# interface type-and-number
Router(config-if)# description description-text
Router(config-if)# ip address ipv4-address subnet-mask
Router(config-if)# ipv6 address ipv6-address/prefix-length
Router(config-if)# no shutdown
```

- It is a good practice to use the **description** command to add information about the network connected to the interface.
- The **no shutdown** command activates the interface.

Configure Router Interfaces Example

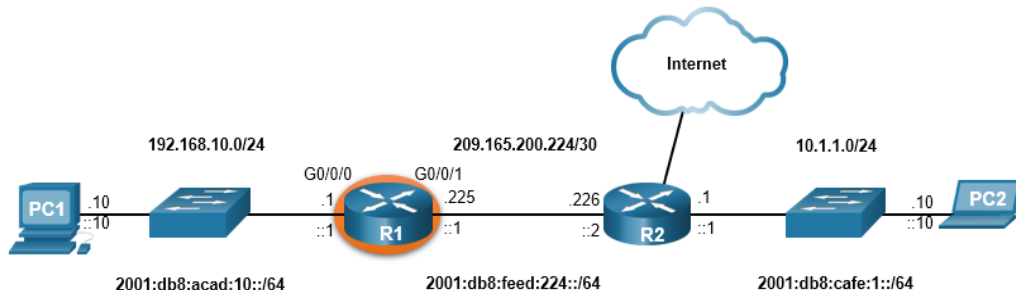
The commands to configure interface G0/0/0 on R1 are shown here:



```
R1(config)# interface gigabitEthernet 0/0/0
R1(config-if)# description Link to LAN
R1(config-if)# ip address 192.168.10.1 255.255.255.0
R1(config-if)# ipv6 address 2001:db8:acad:10::1/64
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)#
*Aug 1 01:43:53.435: %LINK-3-UPDOWN: Interface GigabitEthernet0/0/0, changed state to down
*Aug 1 01:43:56.447: %LINK-3-UPDOWN: Interface GigabitEthernet0/0/0, changed state to up
*Aug 1 01:43:57.447: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0,
changed state to up
```

Configure Router Interfaces Example (Cont.)

The commands to configure interface G0/0/1 on R1 are shown here:



```
R1(config)# interface gigabitEthernet 0/0/1
R1(config-if)# description Link to R2
R1(config-if)# ip address 209.165.200.225 255.255.255.252
R1(config-if)# ipv6 address 2001:db8:feed:224::1/64
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)#
*Aug 1 01:46:29.170: %LINK-3-UPDOWN: Interface GigabitEthernet0/0/1, changed state to down
*Aug 1 01:46:32.171: %LINK-3-UPDOWN: Interface GigabitEthernet0/0/1, changed state to up
*Aug 1 01:46:33.171: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1,
changed state to up
```

Configure Interfaces

Verify Interface Configuration

To verify interface configuration use the **show ip interface brief** and **show ipv6 interface brief** commands shown here:

```
R1# show ip interface brief
Interface                IP-Address      OK? Method Status          Protocol
GigabitEthernet0/0/0     192.168.10.1    YES manual up              up
GigabitEthernet0/0/1     209.165.200.225 YES manual up              up
Vlan1                    unassigned      YES unset  administratively down down
```

```
R1# show ipv6 interface brief
GigabitEthernet0/0/0     [up/up]
FE80::201:C9FF:FE89:4501
2001:DB8:ACAD:10::1
GigabitEthernet0/0/1     [up/up]
FE80::201:C9FF:FE89:4502
2001:DB8:FEED:224::1
Vlan1                    [administratively down/down]
unassigned
R1#
```

Configure Interfaces

Configure Verification Commands

The table summarizes show commands used to verify interface configuration.

Commands	Description
<code>show ip interface brief</code> <code>show ipv6 interface brief</code>	Displays all interfaces, their IP addresses, and their current status.
<code>show ip route</code> <code>show ipv6 route</code>	Displays the contents of the IP routing tables stored in RAM.
<code>show interfaces</code>	Displays statistics for all interfaces on the device. Only displays the IPv4 addressing information.
<code>show ip interfaces</code>	Displays the IPv4 statistics for all interfaces on a router.
<code>show ipv6 interfaces</code>	Displays the IPv6 statistics for all interfaces on a router.

Configure Verification Commands (Cont.)

View status of all interfaces with the **show ip interface brief** and **show ipv6 interface brief** commands, shown here:

```
R1# show ip interface brief
Interface                IP-Address      OK? Method Status          Protocol
GigabitEthernet0/0/0     192.168.10.1    YES manual up              up
GigabitEthernet0/0/1     209.165.200.225 YES manual up              up
Vlan1                    unassigned      YES unset  administratively down down
R1#
```

```
R1# show ipv6 interface brief
GigabitEthernet0/0/0     [up/up]
FE80::201:C9FF:FE89:4501
2001:DB8:ACAD:10::1
GigabitEthernet0/0/1     [up/up]
FE80::201:C9FF:FE89:4502
2001:DB8:FEED:224::1
Vlan1                    [administratively down/down]
unassigned
R1#
```

Configure Verification Commands (Cont.)

Display the contents of the IP routing tables with the **show ip route** and **show ipv6 route** commands as shown here:

```
R1# show ip route
< output omitted >
Gateway of last resort is not set
    192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0/0
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0/0
    209.165.200.0/24 is variably subnetted, 2 subnets, 2 masks
C       209.165.200.224/30 is directly connected, GigabitEthernet0/0/1
L       209.165.200.225/32 is directly connected, GigabitEthernet0/0/1
R1#
```

```
R1# show ipv6 route
<output omitted>
C   2001:DB8:ACAD:10::/64 [0/0]
    via GigabitEthernet0/0/0, directly connected
L   2001:DB8:ACAD:10::1/128 [0/0]
    via GigabitEthernet0/0/0, receive
C   2001:DB8:FEED:224::/64 [0/0]
    via GigabitEthernet0/0/1, directly connected
L   2001:DB8:FEED:224::1/128 [0/0]
    via GigabitEthernet0/0/1, receive
L   FF00::/8 [0/0]
    via Null0, receive
R1#
```

Configure Interfaces

Configure Verification Commands (Cont.)

Display statistics for all interfaces with the **show interfaces** command, as shown here:

```
R1# show interfaces gig0/0/0
GigabitEthernet0/0/0 is up, line protocol is up
  Hardware is ISR4321-2x1GE, address is a0e0.af0d.e140 (bia a0e0.af0d.e140)
  Description: Link to LAN
  Internet address is 192.168.10.1/24
  MTU 1500 bytes, BW 100000 Kbit/sec, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive not supported
  Full Duplex, 100Mbps, link type is auto, media type is RJ45
  output flow-control is off, input flow-control is off
  ARP type: ARPA, ARP Timeout 04:00:00
  Last input 00:00:01, output 00:00:35, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/375/0/0 (size/max/drops/flushes); Total output      drops: 0
  Queueing strategy: fifo
  Output queue: 0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    1180 packets input, 109486 bytes, 0 no buffer
    Received 84 broadcasts (0 IP multicasts)
    0 runts, 0 giants, 0 throttles

<output omitted>

R1#
```

Configure Verification Commands (Cont.)

Display IPv4 statistics for router interfaces with the **show ip interface** command, as shown here:

```
R1# show ip interface g0/0/0
GigabitEthernet0/0/0 is up, line protocol is up
  Internet address is 192.168.10.1/24
  Broadcast address is 255.255.255.255
  Address determined by setup command
  MTU is 1500 bytes
  Helper address is not set
  Directed broadcast forwarding is disabled
  Outgoing Common access list is not set
  Outgoing access list is not set
  Inbound Common access list is not set
  Inbound access list is not set
  Proxy ARP is enabled
  Local Proxy ARP is disabled
  Security level is default
  Split horizon is enabled
  ICMP redirects are always sent
  ICMP unreachable are always sent
  ICMP mask replies are never sent
  IP fast switching is enabled
  IP Flow switching is disabled
```

<output omitted>

R1#

Configure Interfaces

Configure Verification Commands (Cont.)

Display IPv6 statistics for router interfaces with the **show ipv6 interface** command shown here:

```
R1# show ipv6 interface g0/0/0
GigabitEthernet0/0/0 is up, line protocol is up
  IPv6 is enabled, link-local address is
FE80::868A:8DFF:FE44:49B0
  No Virtual link-local address(es):
  Description: Link to LAN
  Global unicast address(es):
    2001:DB8:ACAD:10::1, subnet is 2001:DB8:ACAD:10::/64
  Joined group address(es):
    FF02::1
    FF02::1:FF00:1
    FF02::1:FF44:49B0
  MTU is 1500 bytes
  ICMP error messages limited to one every 100 milliseconds
  ICMP redirects are enabled
  ICMP unreachables are sent
  ND DAD is enabled, number of DAD attempts: 1
  ND reachable time is 30000 milliseconds (using 30000)
  ND NS retransmit interval is 1000 milliseconds

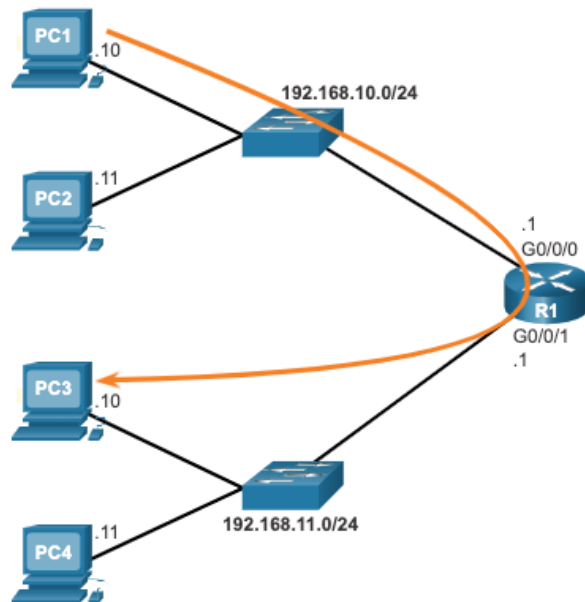
R1#
```

Configure the Default Gateway

Configure the Default Gateway

Default Gateway on a Host

- The default gateway is used when a host sends a packet to a device on another network.
- The default gateway address is generally the router interface address attached to the local network of the host.
- To reach PC3, PC1 addresses a packet with the IPv4 address of PC3, but forwards the packet to its default gateway, the G0/0/0 interface of R1.

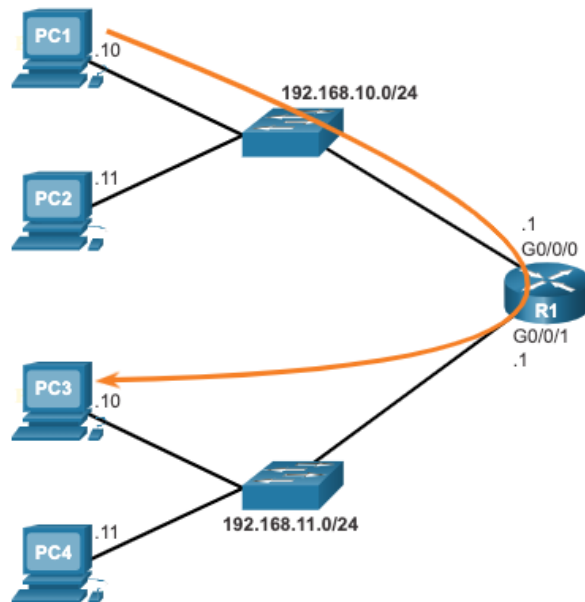


Note: The IP address of the host and the router interface must be in the same network.

Configure the Default Gateway

Default Gateway on a Switch

- A switch must have a default gateway address configured to remotely manage the switch from another network.
- To configure an IPv4 default gateway on a switch, use the **ip default-gateway ip-address** global configuration command.



Lab 3.2C – Connect a Router to a LAN

In this Packet Tracer, you will do the following:

- Display the router information.
- Configure router interfaces.
- Verify the configuration.

